ECON 0100 | Exercise C3 | Corrective Policies

Q1 | Correcting Smoked Toffee

In recent years Hogsmeade has seen a proliferation of candy shops selling smoked toffees. The Demand (marginal benefit) curve and Supply (marginal cost) curve for smoked toffee can be represented by the following relationships.

$$D : P_b = 20 - Q_d \qquad\qquad S : P_s = 2 + Q_s$$

The proliferation of smoked toffee makers meant the town saw a subsequent rise in *smoke*. Wood smoke can cause lung issues. Public health officials have estimated the externality to be equal to carry a cost of 10 Coins to the health of Hogsmeade residents.

What type and size of policy would you propose to eliminate the DWL you identified in Exercise C1? Use a graph and some algebra to illustrate.

Policy Type:	_____	Policy Size:	_____
Buyer's Price (P_b):	_____	Seller's Price (P_s):	_____
Post-Policy Quantity (Q^τ):	_____	Deadweight Loss (DWL):	_____

Q2 | Correcting Yard Flowers

As a small residential town, Hogsmeade has many shops and many homes near the town center. Residential homeowners often plant flowers in their front yards, contributing to the positive atmosphere in town. There are numerous garden stores selling supplies for planting and growing many varieties. The supply and demand relationships for a dozen flowers is given by the following equations, with prices in Coins.

$$D : P = 100 - \frac{1}{2}Q_d \qquad S : P = \frac{1}{2}Q_s$$

Recently, a study found a positive impact on the wellbeing of the community from the flowers planted in the front yards of their neighbors. The impact was estimated to carry a value of 10 Coins per dozen flowers planted.

What type and size of policy would you propose to eliminate the DWL you identified in Exercise C1? Use a graph and some algebra to illustrate.

Policy Type:	_____	Policy Size:	_____
Buyer's Price (P_b):	_____	Seller's Price (P_s):	_____
Post-Policy Quantity (Q^τ):	_____	Deadweight Loss (DWL):	_____